

Saimon Bhuiyan

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[LinkedIn](#), [Scholar](#), [Portfolio](#)

Education

Bachelor of Science in Software Engineering (BSSE)

September 2022 – Present

University of Dhaka

CGPA: 3.86/4.00

Dhaka, Bangladesh

Major Courseworks: Data Structure and Algorithms I & II, Machine Learning, Artificial Intelligence, Database Management System I & II, Object Oriented Programming I & II, Image Processing

Higher Secondary Certificate (HSC)

2019 – 2021

Narsingdi Government College

GPA: 5.00/5.00

Narsingdi, Bangladesh

Secondary School Certificate (SSC)

2017 – 2019

Satirpara K. K. Institution School and College

GPA: 5.00/5.00

Narsingdi, Bangladesh

Awarded Government Scholarship for Outstanding Academic Performance

Research Interest

My research focuses on building safe and trustworthy AI systems. I explore how Explainable AI (XAI) and mechanistic interpretability techniques can be applied to inspect neural representations, evaluate reliability, and address safety risks in AI deployment.

Research Activities

Conference and Workshop Proceedings

- Wasi, A. T., Khan, M. A., Rahman, A., Faisal, W., Saha, S., **Bhuiyan, S.**, Sukanya, M. R., Priti, R. N., Hoque, M. I., Shahriyar, M., Islam, R., Sushil, S., Ridoy, S. Z., Sultan, K. R., Islam, M. S., Ahsan, M. M., & Parvez, M. R. (2026). InvariantBench: Can Large Language Models Exhibit Inherent Reasoning Across Equivalent Transformations? In *Proceedings of the ICML 2026 Workshop on AI4Math*.
- Bhuiyan, S.**, Roni, M. K. H., Sukanya, M. R., Mohsin, M. B., & Sayad, M. (2026). Neuro-Sync Adaptive Compression (NSAC): A Multi-Stage Pipeline for LLM Inference at Low Resource Environment. In *Proceedings of the IEEE International Conference on Information, Research and Advanced Innovations (IRAI) 2026*.

Research Ongoing

- Bhuiyan, S.**, Nahid, S. S., & Hossain, B. M. M. Towards Inherently Interpretable Neural Networks: A Jointly Optimized Unified Axiomatic Co-Hoc Training Framework. (*Manuscript in preparation*).

Research Experience

Computational Intelligence and Operations Laboratory (CIOL)

February 2026 – Present

Research Student

Remote

- Contributed to the *InvariantBench* dataset (accepted at ICML 2026 Workshop on AI4Math) through data curation, manual annotation, and quality assurance for LLM reasoning evaluation.
- Trained in fundamental research practices, focusing on dataset engineering, task formulation, and multi-institutional academic collaboration.

Undergraduate Researcher

2025 – Present

Institute of Information Technology

University of Dhaka

- Jointly Optimized Axiomatic Framework** (*Manuscript in preparation*): Identified core research gaps, conceptualized the full architectural methodology, and conducted empirical experiments under faculty supervision.

- **Neuro-Sync Adaptive Compression (NSAC)** (IEEE IRAI 2026): Served as Project Team Lead; managed cross-functional workflows, directed team execution, and co-formulated the compression methodology and experimental evaluation.

Work Experiences

StreamsTech

November 2025 – May 2026

ML and Generative AI Intern

Onsite

- Engineered an employee working hour tracker utilizing live CCTV footage using CNN architectures.
- Architected high-performance production dashboards and developed Python-based backends integrated with AI agents.

Factoryze

January 2025 – April 2025

Software Engineer Intern

Remote

- Contributed to a full-stack property rental ecosystem for an American client using Next.js.
- Optimized frontend interface layers and collaborated with cross-functional technical teams.

Professional Service & Mentorship

Bangladesh AI Olympiad (BDAIO)

2026

Judge and Problem Setter (Regional Round)

- Designed competitive algorithmic and machine learning problems tailored for testing foundational AI capabilities.
- Evaluated participant submissions and contributed to benchmarking performance metrics for regional selections.

Independent Academic Tutoring

2024-Present

Undergraduate Tutor (International)

- Provided target academic mentoring to undergraduate computer science and engineering students from international institutions including Murdoch University and Curtin University, Australia.
- Delivered foundational and advanced instruction in core software engineering topics, data structures & algorithm and networking.

Skills

- **Core Domains:** Deep Learning, Explainable AI (XAI), Mechanistic Interpretability, Transformer
- **ML Frameworks & Libraries:** PyTorch, Hugging Face (Transformers, PEFT), NumPy, Pandas, Scikit-Learn
- **Languages:** Python, Java, TypeScript, C++, C, SQL, Bash

Honors & Awards

- **Finalist**, IEEE IES Generative AI Challenge 2026.
- **Honorable Mention (AI Category)**, Blockchain Olympiad Bangladesh 2024 (Nominated for International Round) for the project titled "RenewAble Synergy: Optimizing Production, Distribution, and Storage in Renewable Energy Systems Using AI for a Sustainable Future"
- **Participant**, International AI Olympiad (IDSOL) 2024, Amsterdam, Netherlands.
- **1st Place**, East West University IT Olympiad 2024.
- **4th Place (Sub-District)**, Science Olympiad of 39th National Science & Technology Week 2018

Leadership Experiences

IITSEC (IIT Software Engineer Community)

Executive Member

- Organized major departmental events, spearheaded logistics planning, and facilitated communication channels between university administration and students.

IIT Football Team, University of Dhaka

Assistant Manager & Former Player

- Overseeing squad coordination, operational logistics, and match-day strategies for the team competing in the Inter-Departmental Football Tournament.
- Represented the department on the field as a core player in previous tournament iterations.